Homework 10

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Contents

Problem 1

Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is complete, then Y is closed in X. (Hint: It suffices to show $\overline{Y} \subseteq Y$. Use sequences.)

Discussion & Scratch Work

> The key definitions here come straight from lecture. First, saying that $Y \subseteq X$ is *complete* means that every Cauchy sequence in the subspace (Y, d_Y) converges to some point of Y, where d_Y is the induced metric. Since the induced metric is just the restriction of d to pairs of points in Y, a sequence in Y is Cauchy in (Y, d_Y) exactly when it is Cauchy in (X, d).
>
> The most important theorem from lecture for this problem is the sequential characterization of closure: for a subset $A \subseteq X$,
>
> $$x \in \overline{A} \quad \Longleftrightarrow \quad \text{there exists a sequence } \{a_n\} \subseteq A \text{ such that } a_n \to x.$$
>
> We also proved that every convergent sequence in a metric space is Cauchy, and that limits in a metric space are unique.
>
> So the strategy is very direct. Since the hint says it suffices to prove $\overline{Y} \subseteq Y$, I start with an arbitrary point $x \in \overline{Y}$. By the closure theorem, there is a sequence $\{y_n\} \subseteq Y$ with $y_n \to x$ in X. Because convergent sequences are Cauchy, $\{y_n\}$ is Cauchy. Since Y is complete, that same sequence must converge to some point $y \in Y$. But now the sequence $\{y_n\}$ converges to both x and y in the metric space X, so uniqueness of limits forces $x = y$. Hence $x \in Y$. Since every point of $\overline{Y}$ lies in Y, we get $\overline{Y} \subseteq Y$, and therefore Y is closed.

Proof.
We will show that

$$\overline{Y} \subseteq Y.$$

Since always $Y \subseteq \overline{Y}$, this will imply $Y = \overline{Y}$, and hence Y is closed in X.

So let $x \in \overline{Y}$. By the sequential characterization of closure proved in lecture, there exists a sequence $\{y_n\}_{n \geq 1}$ in Y such that

$$y_n \to x \qquad \text{in } (X, d).$$

Since every convergent sequence in a metric space is Cauchy, the sequence $\{y_n\}$ is Cauchy in (X, d). Because each y_n lies in Y and the induced metric on Y is just the restriction of d, it follows that $\{y_n\}$ is also a Cauchy sequence in the subspace (Y, d_Y).

Now Y is complete by assumption, so every Cauchy sequence in (Y, d_Y) converges to a point of Y. Therefore there exists some $y \in Y$ such that

$$y_n \to y \qquad \text{in } (Y, d_Y).$$

But convergence in the subspace metric means exactly the same distance condition as convergence in (X, d), so in fact

$$y_n \to y \qquad \text{in } (X, d).$$

Thus the sequence $\{y_n\}$ converges in (X, d) to both x and y. By uniqueness of limits in metric spaces, we must have

$$x = y.$$

Since $y \in Y$, it follows that $x \in Y$.

Because $x \in \overline{Y}$ was arbitrary, we conclude that

$$\overline{Y} \subseteq Y.$$

Therefore Y is closed in X. $\square$

Problem 2

Let (X, d) be a metric space. Prove that if $A, B \subseteq X$ are both compact then $A \cup B$ is compact.

Discussion & Scratch Work

> The main definition from lecture that matters here is the definition of compactness: a subset $K \subseteq X$ is compact if every open cover of K has a finite subcover. So for this problem, I think I should start with an arbitrary open cover of $A \cup B$ and try to show that only finitely many of those open sets are needed.
>
> The big thing is that any open cover of $A \cup B$ automatically gives an open cover of A and also an open cover of B, since both sets are contained in $A \cup B$. Because A is compact, I can extract finitely many sets from the cover that already cover A. Likewise, because B is compact, I can extract finitely many sets from the same cover that cover B. Then I simply combine those two finite collections. Their union is still finite, and together they cover both A and B, hence they cover $A \cup B$.
>
> So the proof template is: begin with an arbitrary open cover of $A \cup B$, apply compactness once to A, apply compactness again to B, and then merge the two finite subcovers into one finite subcover of the union.

Proof.
Let $\{U_i\}_{i \in I}$ be an open cover of $A \cup B$. Thus each U_i is open in X and

$$A \cup B \subseteq \bigcup_{i \in I} U_i.$$

Since $A \subseteq A \cup B$, it follows that

$$A \subseteq \bigcup_{i \in I} U_i.$$

So $\{U_i\}_{i \in I}$ is an open cover of A. Because A is compact, there exists a finite set $J_A \subseteq I$ such that

$$A \subseteq \bigcup_{i \in J_A} U_i.$$

Similarly, since $B \subseteq A \cup B$, we also have

$$B \subseteq \bigcup_{i \in I} U_i.$$

So $\{U_i\}_{i \in I}$ is an open cover of B. Because B is compact, there exists a finite set $J_B \subseteq I$ such that

$$B \subseteq \bigcup_{i \in J_B} U_i.$$

Now set

$$J := J_A \cup J_B.$$

Since J_A and J_B are finite, the set J is finite. Moreover,

$$A \cup B \subseteq \left(\bigcup_{i \in J_A} U_i \right) \cup \left(\bigcup_{i \in J_B} U_i \right) = \bigcup_{i \in J} U_i.$$

Thus $\{U_i\}_{i \in J}$ is a finite subcover of $A \cup B$.

Since the open cover $\{U_i\}_{i \in I}$ was arbitrary, we conclude that $A \cup B$ is compact. $\square$

Problem 3

Let (X, d) be a metric space.

(a) Prove that if $A \subseteq B \subseteq X$, A is closed in X, and B is compact, then A is compact. (Hint: Given an open cover of A, we can add A^c to get an open cover of B.)

(b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact. (Hint: Use (a).)

Discussion & Scratch Work

Both parts of this problem seem like applications of the open-cover definition of compactness, together with one of the closure results from lecture. For part (a), we are given that $A \subseteq B \subseteq X$, that A is closed in X, and that B is compact. Since compactness is about open covers, I should begin with an arbitrary open cover of A and try to turn it into an open cover of B. The hint tells me exactly how: because A is closed, its complement A^c is open, so if I add A^c to my cover of A, then every point of B is covered either because it lies in A or because it lies in $B \setminus A \subseteq A^c$. Then compactness of B gives a finite subcover, and when I restrict back to A, the set A^c becomes unnecessary.

For part (b), I want to prove that the intersection of two compact sets is compact. The cleanest way to use part (a) is to observe that if C and D are compact, then by the theorem from lecture each compact set is closed. In particular, C is closed in X, and of course $C \cap D \subseteq D$. Since D is compact, part (a) applies with $A = C \cap D$ and $B = D$. That immediately gives compactness of $C \cap D$.

Proof.
(a) Let $\{U_i\}_{i \in I}$ be an arbitrary open cover of A, where each U_i is open in X and

$$A \subseteq \bigcup_{i \in I} U_i.$$

Since A is closed in X, the complement A^c is open in X. I claim that

$$\{U_i\}_{i \in I} \cup \{A^c\}$$

is an open cover of B.

Indeed, let $x \in B$. If $x \in A$, then since $\{U_i\}_{i \in I}$ covers A, we have $x \in U_i$ for some $i \in I$. If $x \notin A$, then $x \in A^c$. In either case, x lies in one of the sets from $\{U_i\}_{i \in I} \cup \{A^c\}$. Hence

$$B \subseteq \left(\bigcup_{i \in I} U_i \right) \cup A^c.$$

So this is an open cover of B.

Because B is compact, there exist finitely many indices $i_1, \ldots, i_n \in I$ such that

$$B \subseteq U_{i_1} \cup \cdots \cup U_{i_n} \cup A^c.$$

Now intersect both sides with A. Since $A \subseteq B$, we obtain

$$A \subseteq A \cap (U_{i_1} \cup \cdots \cup U_{i_n} \cup A^c)\,.$$

Distributing the intersection gives

$$A \subseteq (A \cap U_{i_1}) \cup \cdots \cup (A \cap U_{i_n}) \cup (A \cap A^c).$$

But $A \cap A^c = \varnothing$, so

$$A \subseteq (A \cap U_{i_1}) \cup \cdots \cup (A \cap U_{i_n}).$$

Since each $A \cap U_{i_k} \subseteq U_{i_k}$, it follows that

$$A \subseteq U_{i_1} \cup \cdots \cup U_{i_n}.$$

Thus $\{U_{i_1}, \ldots, U_{i_n}\}$ is a finite subcover of A. Since the original open cover of A was arbitrary, we conclude that A is compact.

(b) Suppose that $C, D \subseteq X$ are both compact. By the theorem from lecture, every compact subset of a metric space is closed. Therefore C is closed in X.

Now note that

$$C \cap D \subseteq D \subseteq X.$$

Since C is closed in X, the set $C \cap D$ is also closed in D; equivalently, we may simply apply part (a) with

$$A = C \cap D, \qquad B = D.$$

Indeed, $A \subseteq B$, the set $A = C \cap D$ is closed in X as the intersection of two closed sets, and $B = D$ is compact. Therefore, by part (a), the set $C \cap D$ is compact. $\square$

Problem 4

Let (X, d) be a metric space and let $Y \subseteq X$. Prove that if Y is compact, then Y is complete.

Discussion & Scratch Work

> The main definition here is the definition of completeness: a metric space is complete if every Cauchy sequence converges in that space. Since the problem assumes that $Y \subseteq X$ is compact, the idea is to start with an arbitrary Cauchy sequence in Y and show that it must converge to a point of Y.
>
> The theorem from lecture that does the heavy lifting is the sequential compactness result: if a set is compact, then every sequence in that set has a subsequence that converges in the set. We also know from lecture that every convergent sequence is Cauchy, and in any metric space a Cauchy sequence can have at most one possible limit.
>
> So the proof template is very standard. I take a Cauchy sequence $\{y_n\}$ in Y. Because Y is compact, the sequence has a subsequence $\{y_{n_k}\}$ that converges to some point $y \in Y$. Then I use the fact that the whole sequence is Cauchy to show that actually the entire sequence must converge to that same point y. That will prove that every Cauchy sequence in Y converges in Y, which is exactly what it means for Y to be complete.

Proof.
To prove that Y is complete, let $\{y_n\}_{n\geq 1}$ be a Cauchy sequence in the subspace (Y, d_Y). Since the induced metric d_Y is just the restriction of d to Y, we may regard $\{y_n\}$ simply as a Cauchy sequence of points in Y.

Because Y is compact, every sequence in Y has a subsequence that converges to a point of Y. Therefore there exists a subsequence $\{y_{n_k}\}_{k\geq 1}$ and a point $y \in Y$ such that

$$y_{n_k} \to y.$$

We claim that in fact the whole sequence $\{y_n\}$ converges to y.

Let $\varepsilon > 0$. Since $\{y_n\}$ is Cauchy, there exists $N_1 \in \mathbb{N}$ such that

$$d(y_n, y_m) < \frac{\varepsilon}{2} \qquad \text{for all } n, m \geq N_1.$$

Since $y_{n_k} \to y$, there exists $N_2 \in \mathbb{N}$ such that

$$d(y_{n_k}, y) < \frac{\varepsilon}{2} \qquad \text{for all } k \geq N_2.$$

Choose $k \geq N_2$ large enough so that also $n_k \geq N_1$. Now let $n \geq N_1$. By the triangle inequality,

$$d(y_n, y) \leq d(y_n, y_{n_k}) + d(y_{n_k}, y).$$

Since $n \geq N_1$ and $n_k \geq N_1$, the Cauchy condition gives

$$d(y_n, y_{n_k}) < \frac{\varepsilon}{2},$$

and since $k \geq N_2$, the convergence of the subsequence gives

$$d(y_{n_k}, y) < \frac{\varepsilon}{2}.$$

Therefore

$$d(y_n, y) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since this holds for all $n \geq N_1$, we conclude that

$$y_n \to y.$$

Because $y \in Y$, the sequence $\{y_n\}$ converges in Y.

Thus every Cauchy sequence in Y converges to a point of Y, so Y is complete. $\square$